

ROOT SHEET – NO PARTS BY DESIGN (sheets.md s1).

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The ONLY inter-sheet signal net is /SW (the drain node, 142.5 V pk, 9.17 A rms). Its bare /SW spelling comes from the root-local labels written by add_sheet, and six constraints.json entries depend on it. Both 'stage' and 'tank' expose it, which is what keeps the root label connected – a root label with only ONE sheet pin raises label_dangling in KiCad 10.

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GND / +40V / +5V are global POWER SYMBOLS inside the sheets and are deliberately absent here. All three PWR_FLAGS live on 'hk'; a second flag anywhere else collides power_out <-> power_out.

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Every other net is sheet—internal on purpose and carries its /<sheet>/ prefix. Do not 'tidy' them into bare names.

ai-ee Sheet: / File: rf-de-20m.kicad_sch Title: rf-de-20m: root - 20 MHz Class E GaN stage, 200 W into 50		
Size: A3	Date: 2026-08-07	Rev: 1
KiCad E.D.A. 10.0.3		Id: 1/4





EPC2019 PIN 7 IS SUBSTRATE AND IS TIED TO SOURCE. The datasheet states it three times. It is NOT expressible in the footprint – this schematic is the only place it is enforced. All SOURCE pins (2/4/6/7) -> GND; both DRAIN pins (3/5) -> /SW.

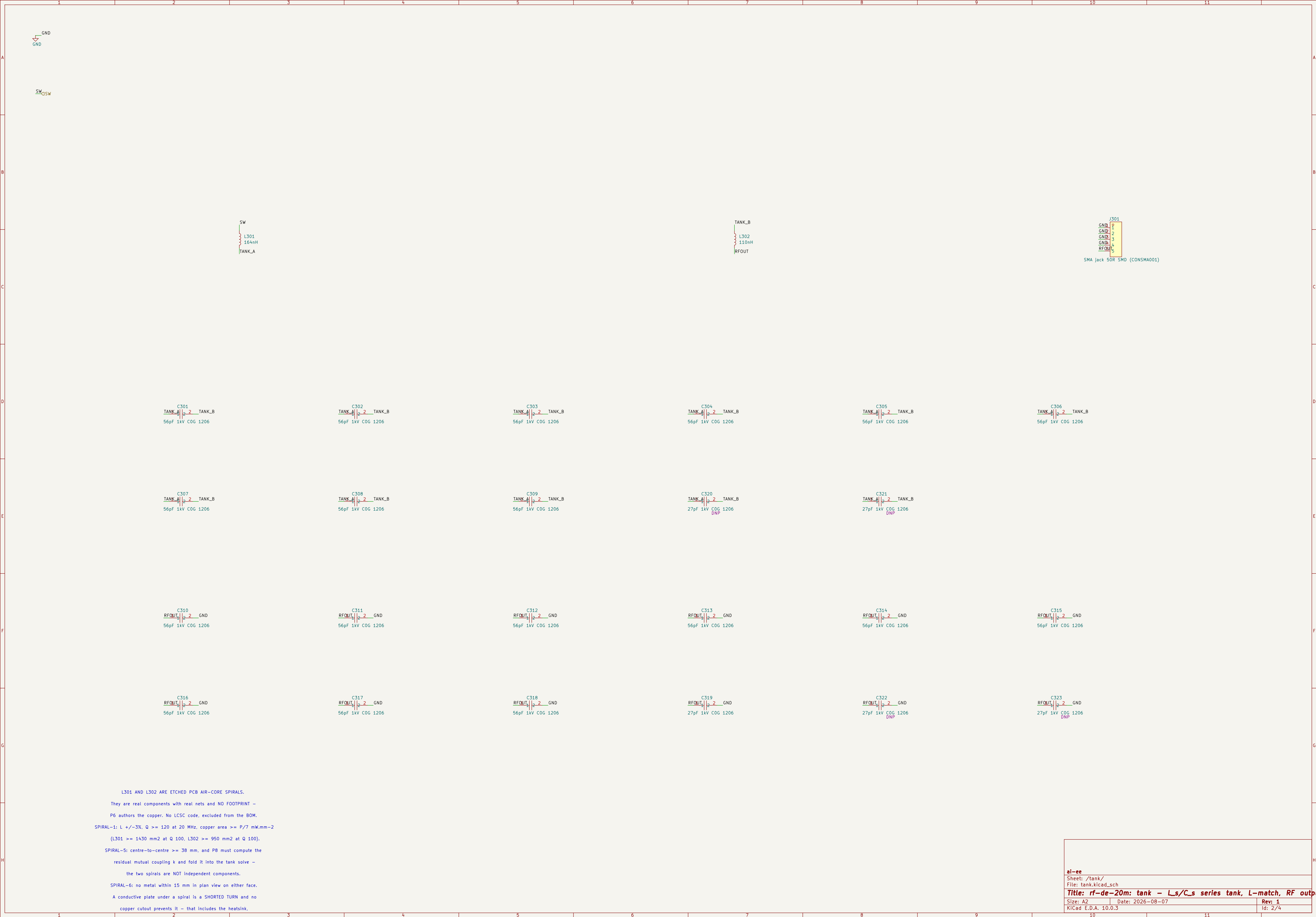
OUTH AND OUTL ARE SEPARATE OUTPUTS. OUTH sources only and floats when idle; OUTL sinks only and is held low otherwise. NO steering diode is needed or wanted. TI'S >=2 OHM MANDATE IS AT THE PIN, AND TWO FETs SHARE EACH PIN, so each branch must be >= 2 x 2 = 4 ohm. ONE 4R7 0805 per leg per FET -> 4.7/2 = 2.35R at OUTH and at OUTL (2.33R at -1%), peak drive 2.13 A. Do NOT parallel these back down. IN- IS TIED TO GND, DRIVE GOES TO IN+. Truth table: only (IN-=L, IN+=H) drives the gate high. IN- has an internal pull-UP to VDD, so floating it holds the stage OFF; IN+ has an internal pull-DOWN, so an open drive input fails SAFE. TI's ground-bounce alternative (drive IN-, IN+ to VDD) is NOT used: it inverts, and duty cycle is the primary ZVS trim.

THE DRIVE INPUT IS DC-COUPLED – RULING DB. Do not add a series blocking capacitor 'for safety': AC coupling's DC restore pins the average at the bias point and at D=0.4 both logic levels sit above the 1.4 V threshold, so the FETs never turn off. Generator MUST be unipolar 0 to +5 V – the input abs max is -0.3 V to VDD+0.3 V, so +/-2.5 V bipolar violates it (OPEN-1).

C203-C206 ARE 56 pF, NOT 33 pF. No 33 pF part exists on this BOM; parts.json maps these sites onto the same 56 pF 1 kV COG part as the tank banks. Populate count re-solved: 2 x 56 = 112 pF external, giving 316 + 112 = 428 pF against a 403-449 pF requirement – mid-band. C205/C206 are DNP headroom; the bank EMPTIES for a max-Coss pair, which is the point of it existing.

GATE LOOPS: aim for <= 0.48 nH per FET, matched +/-0.1 nH, geometrically MIRRRORED about the U201 axis. The CRITICAL-DAMPING budget is now 1.70 nH (EPC WP008 Eq.1 at R = 4.7 + 0.4 + 0.75 = 5.85 ohm, C_GS 199 pF) – 3.6x looser than the 0.48 nH solved at the old 3.1 ohm, which is what pays for the 0805 bodies. Keep aiming at 0.48 nH. Do NOT length-match electrically – FR4 is 6.7 ps/mm and skew is benign in a soft-switched topology; matching damps the differential mode and equalises sharing. L201 + L202 ARE ONE CHOKE, IN SERIES. 0.94 uH total – the architecture's own pre-authorised escape. A smaller single choke is NOT authorised: it breaks the Ideal Class E equations and RAISES the required shunt capacitance.

NO PROTECTION PARTS. No TVS on /SW, no gate clamp, no negative bias. Owner-acknowledged at P0 Q11. The 1.40x derate on a 200 V part is protected by POWER LOOP AREA at P6/P7. FB201 IS A FILTER, NOT PROTECTION. It is not a filter. +5V IS SPLIT AT FB201: +5V_SOURCE feeds the bead, +5V_DRV feeds U201. C213 is not a filter. C213 only. That contains the 10.03 uH. the 72-100 mA of gate charge at 20 MHz must return inside the local 1.1 uF, not across



L301 AND L302 ARE ETCHED PCB AIR-CORE SPIRALS.
They are real components with real nets and NO FOOTPRINT –
P6 authors the copper. No LCSC code, excluded from the BOM.
SPIRAL-1: L +/-3%, Q >= 120 at 20 MHz, copper area >= P/7 mW.mm-2
(L301 >= 1430 mm2 at Q 100, L302 >= 950 mm2 at Q 100).
SPIRAL-5: centre-to-centre >= 38 mm, and PB must compute the
residual mutual coupling k and fold it into the tank solve –
the two spirals are NOT independent components.
SPIRAL-6: no metal within 15 mm in plan view on either face.
A conductive plate under a spiral is a SHORTED TURN and no
copper cutout prevents it – that includes the heatsink,
brackets, standoffs and fasteners.

Zone B carries no plane on In1/In2/B.Cu by construction, and a